

Software Testing & Security Checking

Fall 2023, EE 5191

Syllabus

王凡 (Farn Wang)

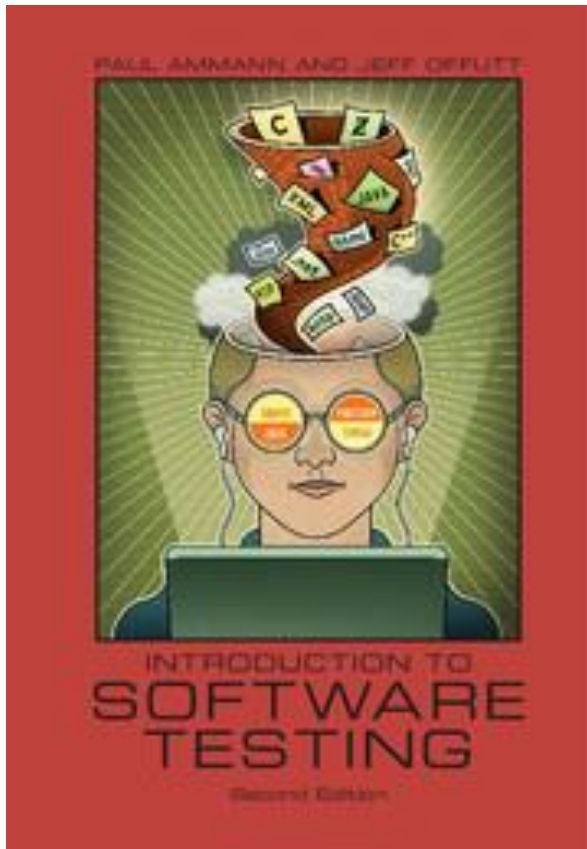
farn@ntu.edu.tw

<http://cc.ee.ntu.edu.tw/ST>

國立臺灣大學 電機工程學系

Textbook

Introduction to Software Testing



Paul Ammann and Jeff Offutt

related materials in

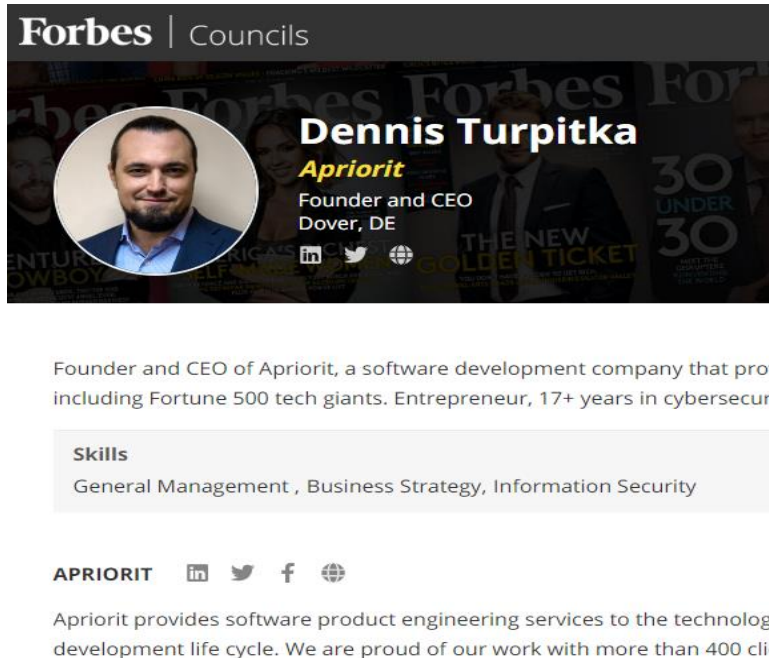
<http://cs.gmu.edu/~offutt/softwaretest/>

Class Information

- 課號：EE 5191 Credits: 3
Fall 2023, Wednesday, 9:10am-12:00noon
- Classroom: BL 112
- 王凡 教授
博理館616，02-33663602
farn@ntu.edu.tw
<http://cc.ee.ntu.edu.tw/~farn/courses/ST>
- 助教：黃浩然
博理館618
r11921104@ntu.edu.tw
Office hour: Wednesday 12:00noon -13:00pm



Status in the Industry



APP測試成本居高不下

- 30-35%研發成本：
具圖形介面APP
- 35-50%：如果還有server，
- 持續高漲中

- Software QA & security is a highly visible and fast-growing sector of software testing.
- Taiwan's high-tech companies do not think that they should spend money in software testing.
 - Taiwan's industrial strength in OEM and ODM

產業分析

軟體品管預算膨脹迅速

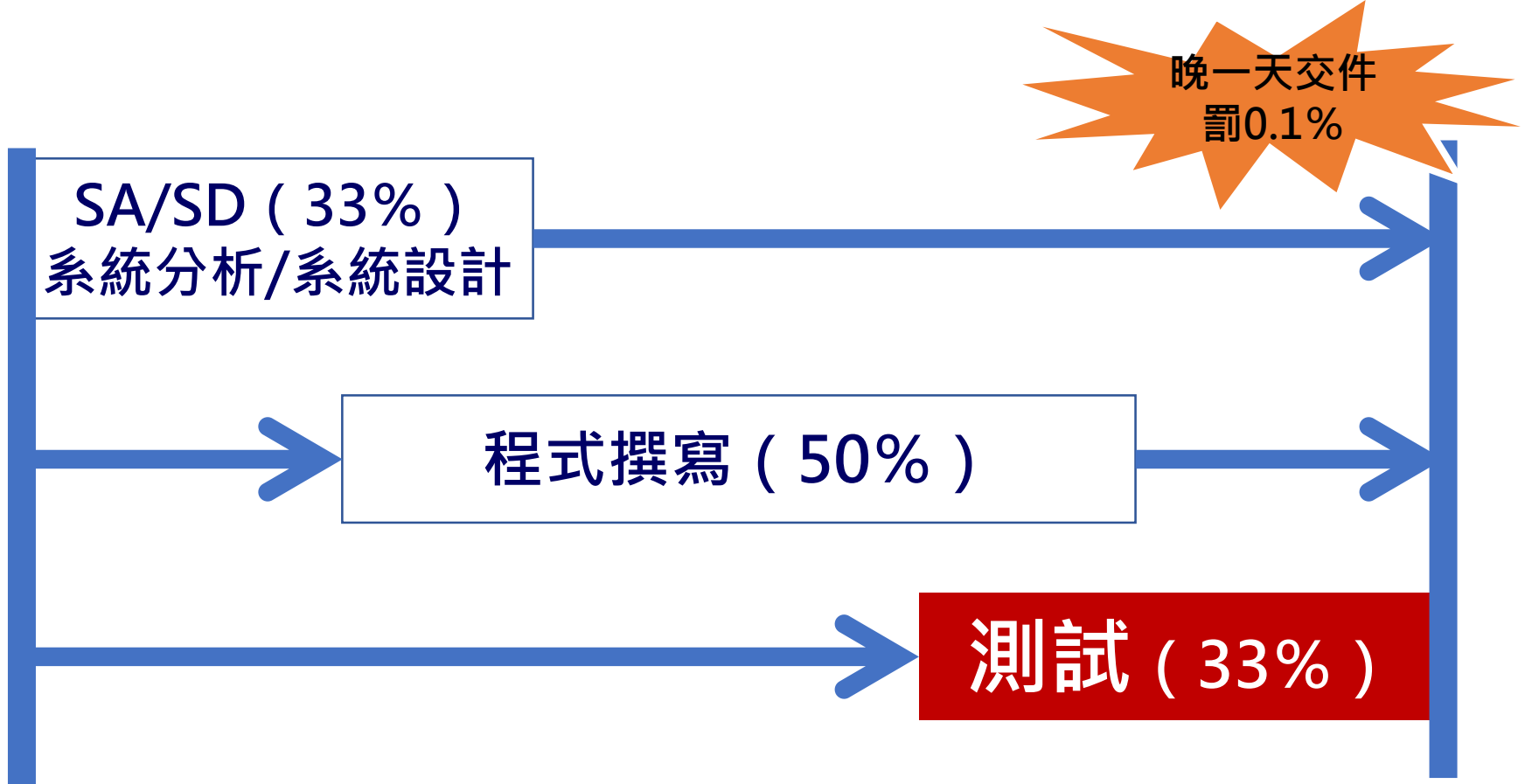
- 目前一般軟體開發計畫，30%以上成本與時間都花在測試與品管。
- 若能降低10%軟體品管成本，就可提昇公司5%獲利。

軟體品管技術交流不易

- 空有天下優良軟體工程師，卻只能進行手工測試。
- 目前一般透過網路廣告，測試需求顧客與軟體測試服務公司個別接洽！
- 底層結構（infrastructure）把持在少數測試工具公司！

產業分析

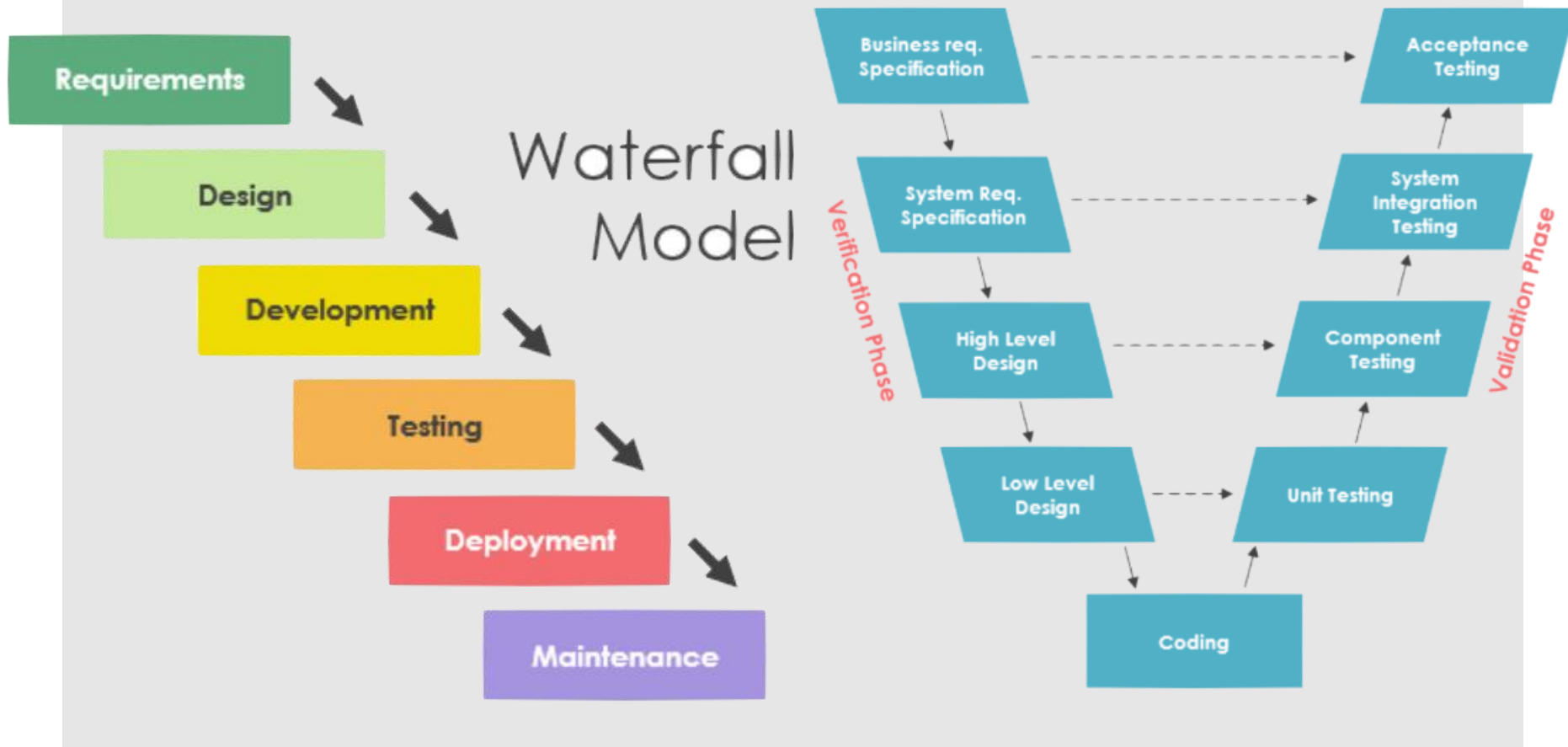
產品差異化訴求 - 軟體開發業者的核心痛點



Software Process Models

現有軟體開發模型測試佔開發預算30-50%

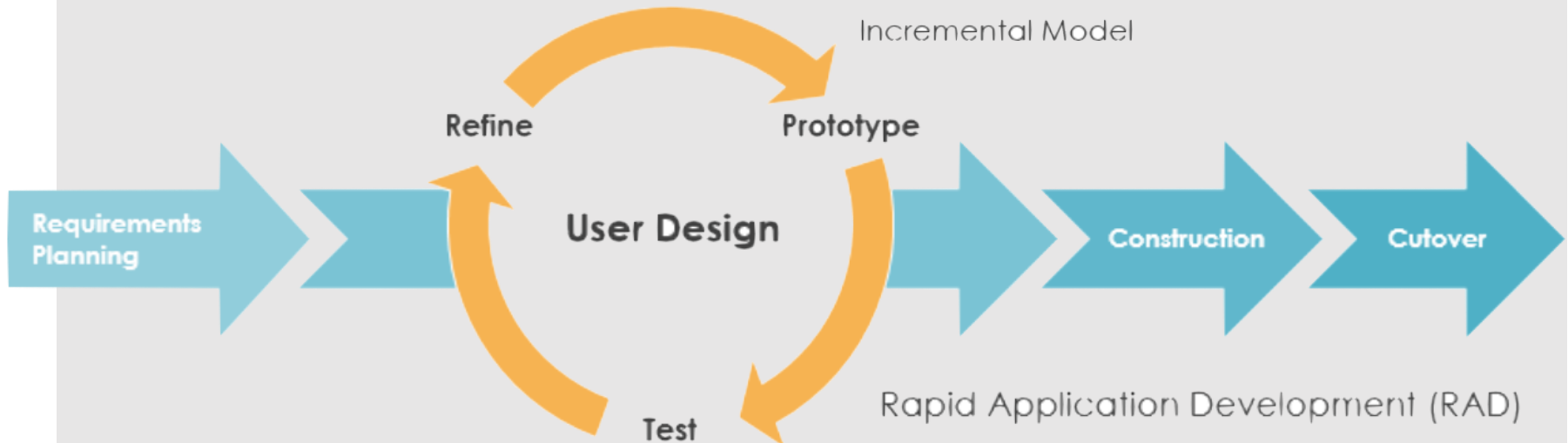
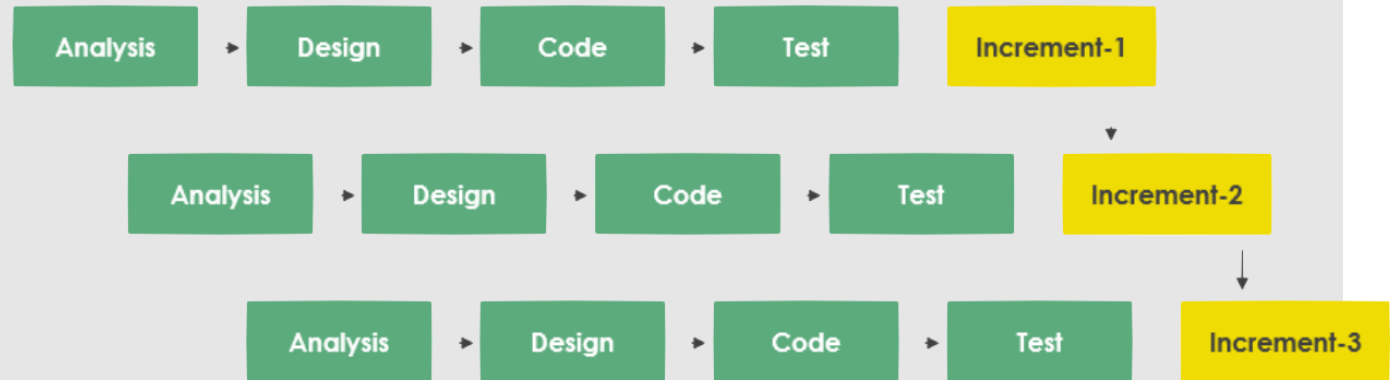
尚不考慮implementation/design/coding階段之測試成本



Software Process Models

現有軟體開發模型測試佔開發預算30-50%

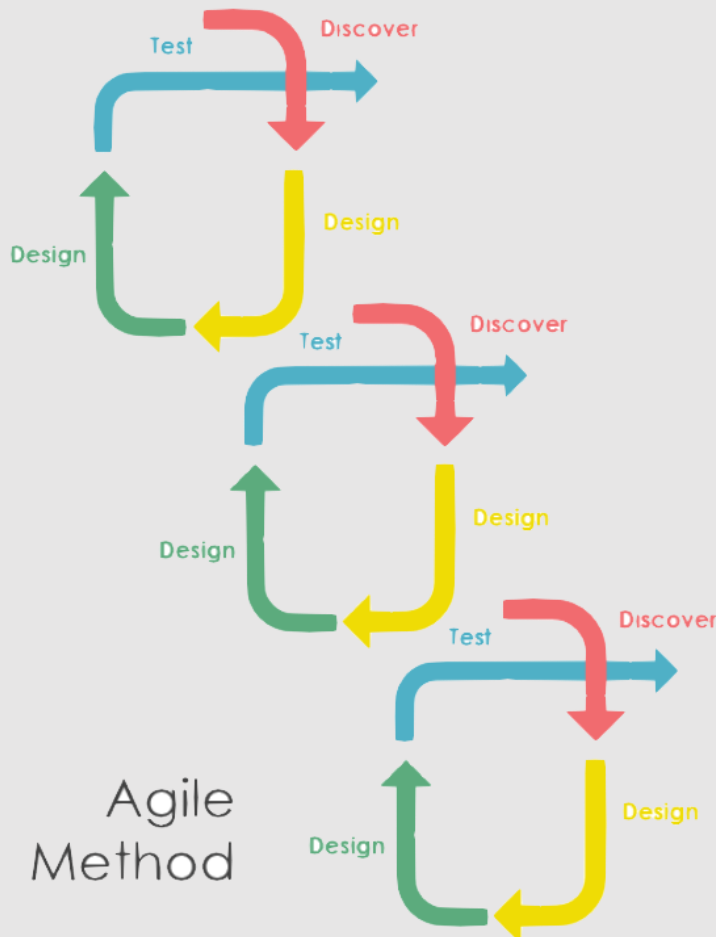
尚不考慮implementation/design/coding階段之測試成本



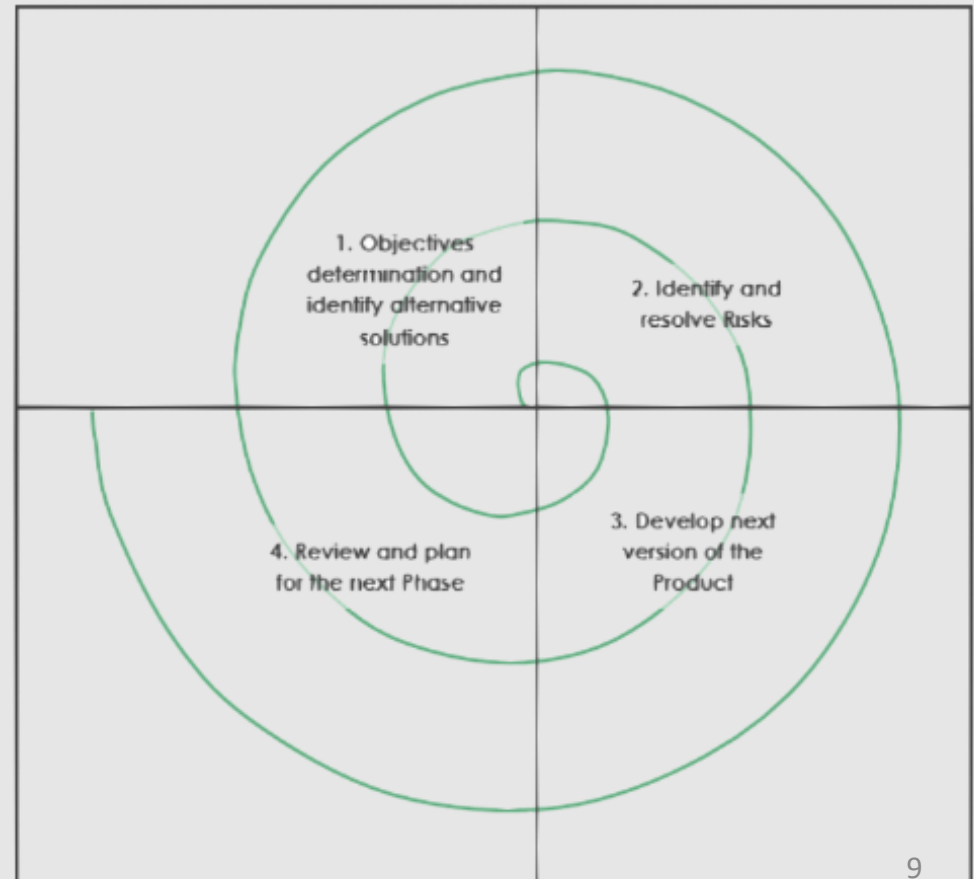
Software Process Models

現有軟體開發模型測試佔開發預算30-50%

尚不考慮implementation/design/coding階段之測試成本



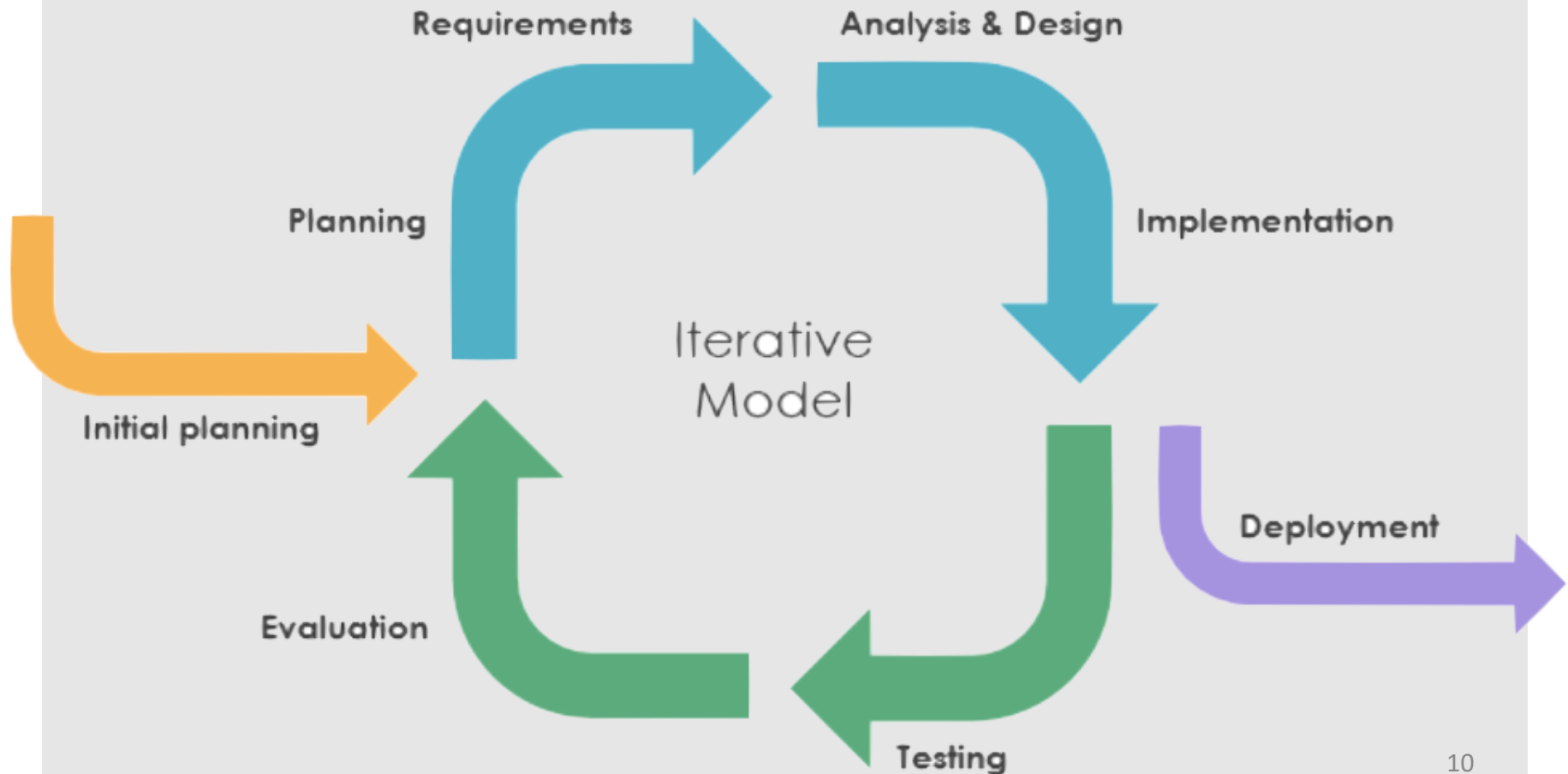
Spiral Model



Software Process Models

現有軟體開發模型測試佔開發預算30-50%

尚不考慮implementation/design/coding階段之測試成本



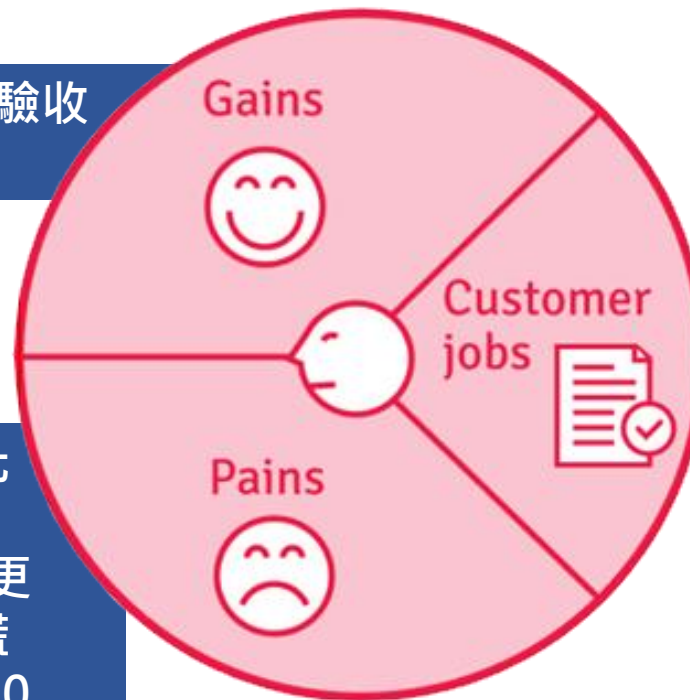
產業分析

產品差異化訴求 - 客戶素描 1

顧客素描 (XXXX科技股份有限公司)

- 準時通過客戶驗收
- 提昇產品形象

- 全人力，沒自動化
- 人力不穩定/不足
- 系統/測試需求變更
- 客戶測試不過恐慌
- 遲交每日罰1/1000



30+人

- 與電信公司作聯合資訊服務
- 或自行經營資訊服務

無專門QA/測試人力
專案可能有2/3名測試人力，也負責程式

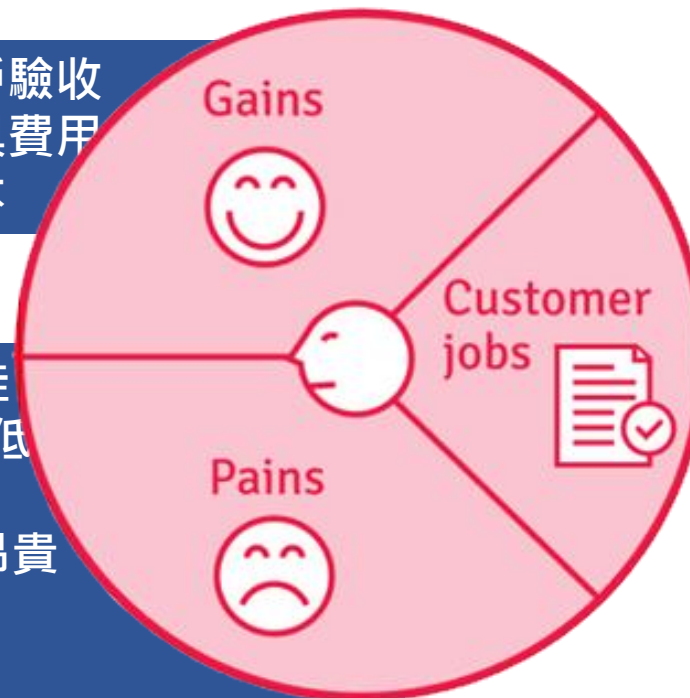
產業分析

產品差異化訴求 - 客戶素描 2

顧客素描（飛象資訊股份有限公司）

- 準時通過客戶驗收
- 降低測試工具費用
- 降低人力成本

- 自動化工具效果不佳
- Selenium投資回饋低
- 人力不穩定/不足
- 自動掃瞄工具授權昂貴
- 系統/測試需求變更
- 客戶測試不過恐慌
- 遲交每日罰1/1000，最高達20%



12人

- 資訊整合服務
- 資安掃瞄服務

一個QA支援三個PM

Status in the Industry

Functional Testing

- Unit testing
- Integration testing
- End-to-end testing
- Smoke testing
- Sanity testing
- Regression testing
- Acceptance testing
- Interface testing
- White box testing
- Black box testing

Status in the Industry

Non-functional Testing

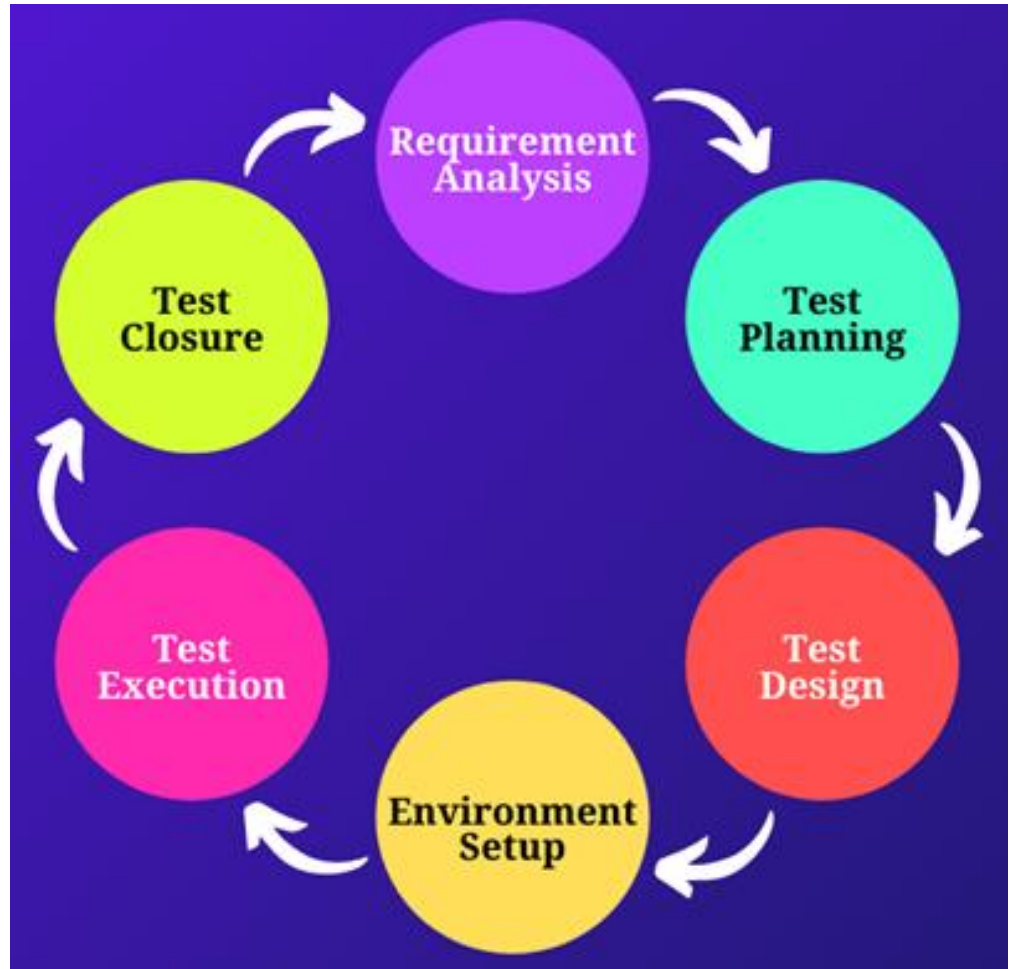
- Performance testing
- Security testing
- Load testing
- Failover testing
- Compatibility testing
- Usability testing
- Scalability testing
- Volume testing
- Stress testing
- Accessibility testing
- Alpha testing
- Beta testing
- Maintainability testing
- Compliance testing
- Efficiency testing
- Reliability testing
- Endurance testing
- Disaster recovery testing
- Localization testing
- Internationalization testing
- Ad-hoc testing

Steps in a Testing Project

現有自動化解決方案

幾乎都著重在

- Test Execution
- Environment Setup



Status of the industry

AUTOMATION TESTING MARKET

Size By Endpoint Interface, 2020 - 2030 (USD Million)

- 預估 2030 年全球E2E SW Testing 市場規模約為 US\$19B
- 若能取得 2.5% 市佔率，營收規模可達到 US\$475M



Status of the industry

年營收

M : 1 百萬美元Revenue

- QualityLogic, <https://www.qualitylogic.com/> (9.7M)
- V-Cubed Solutions, <https://www.v-cubed.net/> (27M)
- Value Stream Engineering, <http://www.valvestreamengineers.com/> (3.8M)
- IT Automation LLC, <http://www.itautomation.com/> (8.8M)
- NeoTys, <https://www.neotys.com/> (8.8M)
- Parasoft, <https://www.parasoft.com/> (55M)
- Performancelab, <https://www.performancelab.com.br/home> (25.3M)
- ScienceSoft, <https://www.scnsoft.com/> (22M)
- Symbio, <https://www.symbio.com/> (150M)
- tenio.io, <https://tenon.io/>

~310.4 Millions
Avg. 34.5 Millions

Background

from the Boss' perspective

Once upon a time in a QA meeting when MS hired 10000 testers,

“I know I pay a lot of testers, but I really have no idea what they do.”



<https://softwaretesting.cioreview.com/cxoinsight/the-cost-of-software-testing-nid-27746-cid-112.html>

Background

from Taiwan Boss' perspective



Status in the Industry

Q #35) How many test cases have you automated per day?

Answer: Well, the number depends on the complexity of the test cases. When the complexity was limited, I was able to automate **5 to 6 test cases per day**.

Sometimes, I was able to automate **only one test case for complex scenarios**.

I have also broken down my test cases into different components like, take input, do the calculation, verify the output etc.

in case of **very complex scenarios and have taken 2 or more days**.

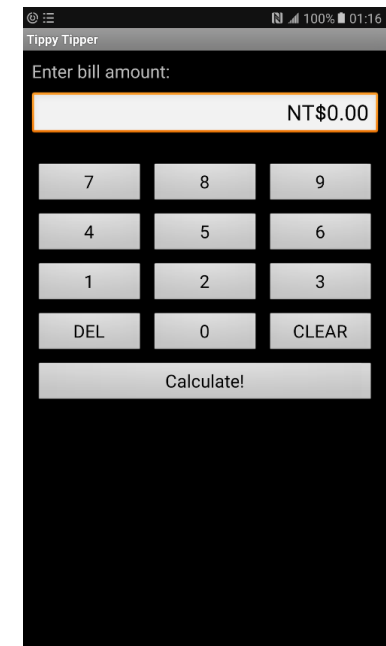
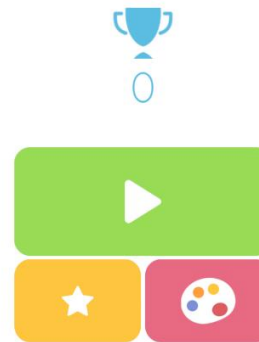


[39 TOP Automation Testing Interview Questions And Answers](https://www.softwaretestinghelp.com/test-automation-interview-questions/)

<https://www.softwaretestinghelp.com/test-automation-interview-questions/>

Status of the industry

- *Basically record/replay and test management.*
- *Test scripts are not reusable across projects.*
- *Need human to design tests*
- *Need human to recognize screens*
- *Need human to understand traces.*



產業分析

solutions nowadays

測試外包

- 適合大型軟體公司
- 只能做system test或user acceptance testing
- IP外洩風險
- Feedback 緩慢

軟體測試自動化工具

- 只幫你record與replay
- Monkey testing (low intelligence)
- Test investment low reusability
- 許多未經整合的測試準備工作

☆☆巨大市場機會☆☆
App智慧型測試工具

產業分析

應用測試難以自動化

- 目前測試自動化成果重複使用性都非常低
- 對一個顧客所製作的成果，幾乎都沒辦法用在另一個顧客。
 - 軟體流程差異性、軟體文件差異性
- 學界的軟體測試成果基本上沒有在工業界應用可能性。

Goal

- Theory in software testing.
- Advanced testing and security-checking tools
- Capability of deploying software testing technology to various companies.
- Basic concepts and skills in security checking.

Lecture Plan

- The first module: foundations
- The second module: Testing criteria
 - Graph testing
 - Logic testing
 - Input space testing
 - Syntax testing
- Third module: practical consideration.
- Criterias for new app types
- Security checking overview.

Pre-requisites

- Basic Computer Concepts
- C/C++, Java programming
- Python
- Data-Structures
- Algorithms

Evaluation

- Final exam (40%) , Dec. 20, 2023
 - 20% of final exam scores will be for questions on selected midterm reports.
- Mid-term report (10%), deadline on Nov. 1, 2023.
 - Deadline for midterm presentation registration: Sept. 20
 - Confirmation of registration: Sept. 27
- 3 projects (50%)
 - CppUnit and GOV for unit testing (15%), due by Oct. 18, 2023.
 - Selenium for web-testing (15%), due by Nov. 22, 2023.
 - App/Web/Security Testing (20%), due by Dec. 27, 2023.
 - ◆ Experiment with AI

Evaluation

Midterm presentation (10%)

- a main paper
 - must be a regular paper published in ACM/IEEE ICSE 2022-23, ACM FSE 2022-23, IEEE ASE 2022-23, ACM ISSTA 2022-23, ACM PLDI 2022-23, IEEE S&P 2022-23, ACM CCS 2022-2023, or USENIX Security 2022-2023.
 - *No tool paper, no short paper, no affiliated workshop paper.*
- 2 comparison papers related to the main paper.
- Comparing the three papers that you selected in the above.
- Submission including your PPT file and MP4 Video for 10mins.

Evaluation

project 1

White-box unit testing with CPP and GOV. (15%)

- Software under test (SUT) is a github open-source cpp module.
- Must write assertions to evaluate the pass or fail of the test scripts.
- Submission:
 - CPPUnit test modules, test reports, and SUT source code.
 - PPT explains the SUT, your test requirements, your test plan, and your test result with coverage analysis.
- Your score is
$$(\text{line coverage}) \times \log(\text{\#lines} + \text{\#classes}) + 10 * (\text{bugs detected})$$

Evaluation

project 2

Web testing with Selenium. (15%)

- Software under test (SUT) is an open-source web program from github.
- Must write assertions to evaluate the pass or fail of the test scripts.
- Submission:
 - Selenium test scripts, test reports with coverage analysis of the test items.
 - PPT explains the SUT, your test requirements, your test plan, and your test result.
- Your score is
 $(\text{coverage of the test items}) \times \log(\# \text{actionables} + \text{html sizes}) + 10 * (\text{bugs detected})$

Evaluation

project 3

App/web/security testing on an AI test automation platform.
(20%)

- Software under test (SUT) could be a mobile APP or a web program.
 - The test items will be announced.
- Writing a python program as a test robot that generates test reports.
- Submission:
 - A test robot that can be used by other classmates.
 - Online documents of your test robot.
- Your score is based on the rating and reviews of peer classmates.

Videos

The screenshot shows a web browser window with the address bar displaying 'cc.ee.ntu.edu.tw/~farn/courses/ST/2021.Spring/'. The page content includes a title 'Videos of lectures for each week:' followed by a bulleted list of video links for five classes. The Windows taskbar at the bottom shows the date and time as '2021/2/24' and '上午 07:23'.

Farn's Software Testing Class

cc.ee.ntu.edu.tw/~farn/courses/ST/2021.Spring/

Videos of lectures for each week:

- Videos for the 1st class (2021/02/24):
[Syllabus](#), [module 1.1](#),
- Videos for the 2nd class (2021/03/03):
[module 1.2](#), [module 1.3](#), [module 1.4](#),
- Videos for the 3rd class (2021/03/10):
[Module 1.5](#), [Module 1.6 & Assignment of market survey](#), [Module 1.7](#),
- Videos for the 4th class (2021/03/17):
[Module 2.1](#), [Module 2.2](#), [Module 2.3](#)
- Videos for the 5th class (2021/03/24):

Course Schedule (1/2)

1.	9/6	a) Syllabus, b) Chapter 1: introduction
2.	9/13	Chapter 1: introduction
3.	9/20	a) Chapter 2: Graph-based Testing b) Deadline for midterm presentation registration c) Project 1 announcement: unit testing
4.	9/27	a) Chapter 2: Graph-based Testing b) Confirmation of midterm presentation registration
5.	10/4	Chapter 2: Graph-based Testing
6.	10/11	Chapter 3: Logic-based Testing
7.	10/18	a) Chapter 3: Logic-based Testing b) Project 1 deadline c) Project 2 announcement: web-testing
8.	10/25	Chapter 3: Logic-based Testing

Course Schedule (2/2)

9.	11/1	a) Deadline of midterm presentation submission b) Chapter 3: Logic-based Testing
10.	11/8	Chapter 4: Input-Space Partition Testing
11.	11/15	校慶停課
12.	11/22	a) Chapter 5: Mutation Testing b) Project 2 deadline c) Project 3 announcement: app/web RWD testing
13.	11/29	Chapter 5: Mutation Testing
14.	12/6	Symbolic Model-Checking
15.	12/13	Security Checking
16.	12/20	Final exam
17.	12/27	Project 3 deadline and demonstration!